Grey Level Transformation

(a) purpose of grey level transformation

$4.5/5$

- ⇒ To improve the visual appearance of image
- ⇒ To enhance contrast and brightness
- ⇒ To highlight important details

(b)

$$s = 255 - r$$

r = orginal pixel value

s = Transformed pixel value

255 = max

original pixel value	Transformed value
0	255
50	205
100	155
150	105
200	55
255	0

Purpose:

- => enhance image contrast
- => spreads pixel intensities over the range
- => Revels hidden details
- => produced a more balanced intensity

b) ## CDF

Step 1: Given histogram

Intensity	frequency
0	4
1	6
2	10
3	6

Total no of pixels =

$$N = 4+6+10+6 = 26$$

Step 2:

Intensity	frequency	CF	CDF = CF/
0	4	4	0.154
1	6	10	0.38
2	10	20	0.2
3	6	26	0.9

Equalized Intensity Value

$L = 4$

Formula

$$S = (L-1) \times CDF = 3 \times CDF$$

PDF?

Result

original	CDF	3×CDF	equalized
0	0.154	0.462	0
1	0.385	1.155	1
2	0.769	2.307	2
3	1.000	3.000	3

Equalized mapping

$0 \to 0$

$1 \to 1$

$2 \to 2$

$3 \to 3$

Gaussian smoothing

Advantages:

* Reduces random noise
* Preserve edge better than average filter
* produces less image blurring

Apply Gaussian Kernel

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 100 & 120 & 100 \\ 120 & 140 & 120 \\ 100 & 120 & 100 \end{bmatrix}$$

multiple each pixels

$$\frac{1}{16}\Big[(1\times100)+(2\times120)+(1\times100)+(2\times120)+(4\times140)$$

$$(2\times120)+(1\times100)+(2\times120)+(1\times100)\Big]$$

$$= \frac{1}{16}(100+240+100+240+560+240+100)$$

$$= \frac{1920}{16} = 120$$

New center pixel Value

$$\boxed{120}$$

Edge Dection using Sobel operator

a) **Purpose :**

=> Detects objects boundaries

=> highlights important image features

=> Reduces unnecessary image information.

b) **Apply the horizontal sobel mask**

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 20 & 30 \\ 10 & 20 & 30 \\ 10 & 20 & 30 \end{bmatrix}$$

$$(-1\times10) + (0\times20) + (1\times30) + (-2\times10) + (0\times20) + (2\times30) + (-1\times10) + (0\times20) + (1\times30)$$

$$= -10 + 0 + 30 - 20 + 0 + 60 - 10 + 0 + 30$$

$$= 80$$

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Gradient at the centre

$$\boxed{G_x = 80}$$

(5) Contrast stretching

(a) Needs:-

=> improves image contrast

=> enhances visibility of fine

=> produces a clear and sharper image.

(b) Linear contrast:

$$S = \frac{(r - r_{min})(L-1)}{r_{max} - r_{min}}$$

$$S = \frac{(r-50) \times 255}{100}$$

original value		Transformed value
50	$(50-50)\times 255/100 = 0$	0
75	$(75-50)\times 255/100 = 0$	64
100	$(100-50)\times 255/100$	128
125	$(125-50)\times 255/100$	191
150	$(150-50)\times 255/100$	255

Transformed value

50 -> 0

75 -> 64

100 -> 128

125 -> 191

150 -> 255